

ANIKET MUKHERJEE

System Architect | Creative Technologist

Target Roles: Media Systems Architect | Workflow Automation Lead | Solution Architect

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Portfolio: <https://aniketmu86.github.io/systems/> | **LinkedIn:** [linkedin.com/in/aniket86](https://www.linkedin.com/in/aniket86)

EXECUTIVE SUMMARY

For 17 years, I have executed frame-accurate motion design and video editing in high-pressure broadcast environments. After nearly two decades in the timeline, my focus shifted from executing repetitive tasks to designing the systems that eliminate operational friction. I architect AI-powered automation pipelines that replace tedious bottlenecks with scalable, standardized infrastructure—automating the heavy lifting while strictly preserving human-in-the-loop validation to ensure elite quality, accuracy, and brand compliance.

My architectural approach is directly informed by my production background. Managing complex asset dependencies, strict broadcast standards, and multi-layered render pipelines translates perfectly into designing deterministic, robust software automation. My systems enable non-technical teams to deliver high-volume media securely and efficiently, significantly reducing dependency on external tools and vendors.

SYSTEMS LEADERSHIP & ARCHITECTURE PRINCIPLES

- **Motion as Logic:** Translating visual timing and broadcast standards into deterministic rules, presets, and controlled rendering pipelines.
- **Human-in-the-Loop Validation:** Designing automation that accelerates execution without bypassing the professional QA and editorial judgment required for high-stakes client deliverables.
- **Secure Ecosystem Deployment:** Building standalone, lightweight systems designed to run without installation in air-gapped and enterprise-constrained environments.
- **Prompt-Driven Architecture:** Decomposing complex workflows into precise logic modules to guide LLM-assisted development of Python/FFmpeg-based systems.

CORE STACK

- **Creative Execution:** Advanced Motion Design, Broadcast Graphics, Adobe After Effects, Premiere Pro, AI-Assisted Workflow Automation.
- **Architecture & Logic:** LLM-Assisted Engineering (Claude/Gemini), Workflow Mapping, Logic Decomposition, Human-in-the-loop Validation.
- **Backend & Processing:** Python 3.12, JavaScript (ExtendScript / RegEx), FFmpeg, Native PyAV, OpenCV.
- **UI & Deployment:** CustomTkinter, Nuitka, PyInstaller (Standalone Enterprise Compilation).

FLAGSHIP PLATFORMS & SYSTEM ARCHITECTURE

Subtitle Studio v5 | The Air-Gapped Production Engine

Architected a zero-dependency standalone suite replacing fragmented video processing and "Shadow IT" converters with a unified, offline-first pipeline.

- **Rapid Development Lifecycle:** Accelerated product development, independently architecting, testing, and compiling the entire v5 suite within a 30-day window using an optimized, LLM-assisted engineering pipeline.
- **Architecture:** Orchestrated a Python-based pipeline integrating native PyAV, FFmpeg demuxing, and a custom premium audio path. Solved complex Nuitka packaging conflicts to ensure 100% portability.
- **The Strategic Pivot:** Replaced commoditized transcription engines with local AI vocal isolation to solve the primary remaining manual audio bottleneck in enterprise environments.
- **Impact:** Democratized branded subtitle generation across the organization; delivered secure, air-gapped deployment with zero IT overhead.

Dynamic Render Pipelines | After Effects Orchestration

Developed an ecosystem of custom ExtendScript (.jsx) architectures that bypass the standard UI to automate complex physics, data parsing, and project lifecycle management directly within the After Effects DOM.

- **Procedural Karaoke Engine:** Deployed a data-driven pipeline that ingests and cross-references dual-SRT data streams. Automatically generates complex, word-level synchronized typography animations completely independent of asset duration, reducing manual timeline animation time by 95%.

- **Automated Vector Rigging Pipeline:** Engineered an LLM-assisted workflow that parses raw SVG coordinate data to automatically generate complex ExtendScript rigging logic. Instantly applies strict mechanical hierarchies and rotational expressions to 2D assets (e.g., beam balances) directly within the application.
- **Asset Dependency & Lifecycle Tracker:** Developed a custom dockable UI panel that dynamically traverses complex project hierarchies. Automatically cross-references and tags footage usage across all nested compositions, eliminating project bloat, optimizing server storage, and standardizing file handoffs between motion designers.

EXPERIENCE

PwC Global L&D (SDC Kolkata)

Senior Associate & Studio Lead | 2024 — Present

- **Creative Leadership & Execution:** Directing high-stakes creative production, managing direct client communications, and driving the hands-on execution of enterprise motion design projects.
- **Workflow Standardization:** Engineered scalable production templates and foundational workflows to ensure strict brand compliance and streamline high-volume delivery across global media teams.
- **Strategic AI Implementation:** Conducted comprehensive assessments of enterprise Generative AI capabilities. Pivoted from basic generation to AI-assisted system development—architecting and deploying proprietary tools (including Subtitle Studio v5) that eliminate manual operational bottlenecks and completely bypass external vendor dependencies.

Creative Lead UK Video & Animation | 2021 — 2023

- Led production pipelines delivering 300+ global assets annually.
- Managed and scaled a 12-member cross-functional team.
- Contributed to winning In-House Agency of the Year 2021 (IHAF).

Senior Motion Designer & Pipeline Architect | 2017 — 2021

- Built foundational workflows, templates, and production systems.
- Developed proprietary media assets, eliminating external licensing costs.

CNN-News18 (Network18)

Broadcast Graphic Designer | 2015 — 2016

- Delivered real-time graphics in a 24/7 zero-error broadcast environment.

Focus Bangla | Bent Cigarette Productions

Motion Designer | 2008 — 2015

- High-volume broadcast motion design and production.

AWARDS & EDUCATION

Awards

- Quality Perfectionist 2022 (PwC)
- In-House Agency of the Year 2021 (IHAF) — UK Video
- "There is no Box" Innovation Award 2020 (PwC)

Education

- Bachelor of Media Science (2005-2008) — Pailan College of Management & Technology